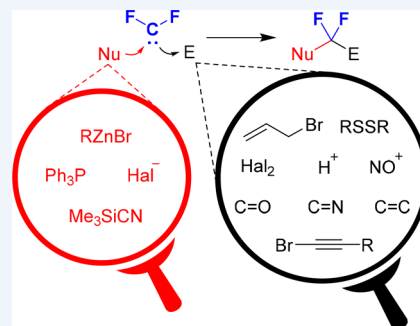


# Difluorocarbene as a Building Block for Consecutive Bond-Forming Reactions

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**CONSPECTUS:** Compounds containing a difluoromethylene unit have gained increasing attention due to their utility in drug design. Classic methods for the synthesis of these compounds rely on either harsh deoxofluorination reactions or laborious functional group manipulation sequences. In 2013, we proposed a method for assembling *gem*-difluorinated molecules from a difluorocarbene, a nucleophile, and an electrophile. In this process, a difluorocarbene can be considered an equivalent of a bipolar  $\text{CF}_2$  unit. Performing consecutive bond-forming reactions by sequential attachment of a nucleophile and an electrophile to a difluorocarbene provides the opportunity for the synthesis of a wide variety of organofluorine compounds. Silicon reagents were the most effective sources of the difluoromethylene fragment, and among them (bromodifluoromethyl)trimethylsilane ( $\text{Me}_3\text{SiCF}_2\text{Br}$ ) is the reagent of choice. Mildly basic activators such as HMPA, DMPU, bromide and acetate ions can initiate the decomposition of the silane with concomitant generation of a difluorocarbene. Organozinc reagents can be employed as nucleophiles, and the  $\text{CF}_2$  fragment can insert into the carbon–zinc bond. Primary and secondary benzyl and alkyl organozinc compounds work well. Generally, organozinc reagents tolerate a variety of functional groups. The resulting fluorinated organozinc species can be coupled with heteroatom- or carbon-centered electrophiles. Halogenation of the carbon–zinc bond leads to compounds with bromo- or iododifluoromethyl fragments, which are difficult to access by other means, whereas protonation of that bond generates a valuable difluoromethyl group. Despite the decrease in the reactivity of the carbon–zinc bond caused by the adjacent fluorines, organozinc compounds can effectively participate in copper-catalyzed cross-couplings with allylic and propargyl halides, 1-bromoalkynes, and *S*-acyl dithiocarbamates. Difluorocarbene can be inserted into the carbon–silicon bond of trimethylsilyl cyanide, and the resulting silane can react with aldehydes and imines to furnish difluorinated nitriles. Interactions of difluorocarbene with heteroatom nucleophiles, such as phosphines or halide ions, are reversible, but the adduct can be trapped by an electrophile. The use of halide ions allows the direct nucleophilic bromo- and iododifluoromethylation of aldehydes and iminium ions. The combination of triphenylphosphine with difluorocarbene generates a difluorinated phosphorus ylide, which can interact with a wide range of  $\pi$ -electrophiles (aldehydes, ketones, acyl chlorides, azomethines, and Michael acceptors) to provide *gem*-difluorinated phosphonium salts. In the latter species, the carbon–phosphorus bond can be readily cleaved under basic conditions, affording the difluoromethylation products. Primary products resulting from three-component couplings can subsequently be used for further transformations. Single-electron reduction of carbon–phosphorus or carbon–iodine bonds can be conducted under photocatalytic conditions to generate *gem*-difluorinated radicals. These radicals can be trapped by silyl enol ethers leading to  $\beta,\beta$ -difluorinated ketones as the primary products. Fluorinated radicals can also undergo intramolecular attacks adjacent to an aromatic ring or a double bond.

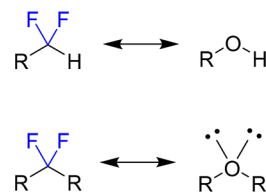


## 1. INTRODUCTION

Over the past 10 years, organofluorine compounds have attracted increasing attention<sup>1</sup> primarily due to their widespread applicability in medicinal chemistry and related fields.<sup>2</sup> Indeed, the number of fluorine-containing drugs and agrochemicals, as well as substances that are in development or clinical trials, is impressive.<sup>3</sup> Typically, in these molecules, the fluorine is present either in the form of a  $\text{CF}_3$ -group or as a substituent, whereas compounds with a difluoromethyl group or a difluoromethylene fragment are encountered less frequently. However, the ability of the  $\text{CHF}_2$  group to serve as a hydrogen bond donor<sup>4</sup> and the isosteric relationship between the  $\text{CF}_2$  fragment and etheral oxygens<sup>5</sup> make difluoromethylated compounds attractive in medicinal research (Scheme 1).

Several approaches for accessing *gem*-difluorinated compounds have been reported<sup>6</sup> (Scheme 2). Deoxofluorination

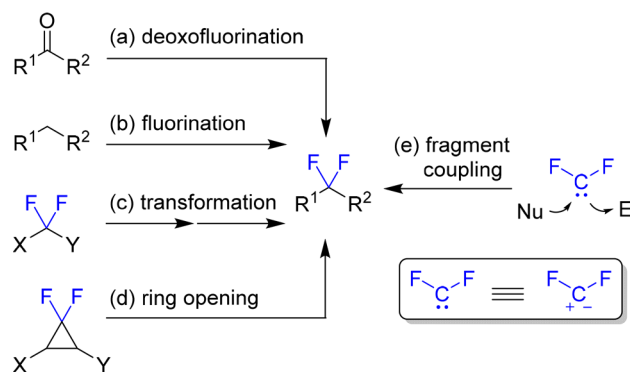
## Scheme 1. Bioisosterism of the $\text{CF}_2$ Fragment



reactions are a general method based on the direct conversion of a carbonyl group into a  $\text{CF}_2$ -fragment<sup>7</sup> (path a). However, this transformation faces major limitations, namely, requiring harsh reaction conditions and hazardous reagents. Direct

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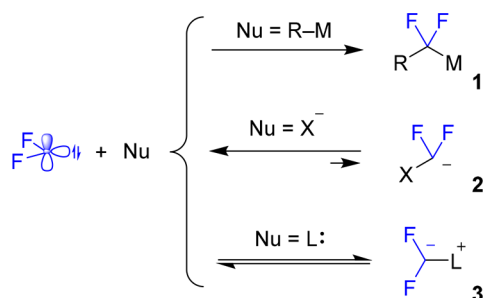
Scheme 2. Synthesis of *gem*-Difluorinated Compounds

fluorination of C–H bonds (path b) has been described based on electrophilic or radical reactions,<sup>8</sup> but this methodology remains to be elaborated. Another approach involves the modification of some readily available fluorinated starting compounds<sup>9</sup> (path c). The benefit of this method is that there is no need for fluorination, while the disadvantage is that the required synthetic sequence may be quite long. The ring-opening of difluorocyclopropanes, which are derived from alkenes, can afford *gem*-difluorinated compounds<sup>10</sup> (path d). Although some examples illustrating this pathway have been reported,<sup>10b–d</sup> this approach is currently of limited scope. Recently, we proposed another method relying on the coupling of three components, namely, a difluorocarbene, a nucleophile, and an electrophile<sup>11</sup> (path e). In this methodology, difluorocarbene can be considered an equivalent of a bipolar  $\text{CF}_2$  unit. If independently varying the nucleophile and electrophile is possible, this method would allow the preparation of a wide variety of difluorinated compounds. A classic way of utilizing difluorocarbene was to difluoromethylate acidic sites,<sup>12</sup> and consecutive bond-forming reactions have not been considered.

## 2. CONCEPT

Difluorocarbene is a singlet carbene,<sup>13</sup> which is intrinsically electrophilic owing to the presence of an empty *p*-orbital (Scheme 3). The interaction of difluorocarbene with a

## Scheme 3. Interaction of Difluorocarbene with Nucleophiles



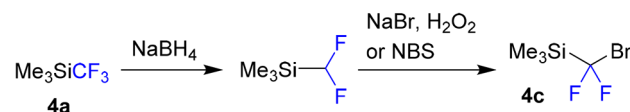
nucleophile can be reversible, and the position of the equilibrium can be shifted. Thus, for organometallic nucleophiles, the formation of reagents **1** is favored due to the formation of a strong C–C bond. On the other hand, with halide ions as nucleophiles, carbanionic species **2** can be unstable and prone to decomposition back to difluorocarbene. Neutral heteroatom nucleophiles can also attack difluorocar-

bene to generate ylides **3**. Structures **1**–**3** are nucleophilic and can be intercepted with suitable electrophiles.

An appropriate source of difluorocarbene is important for the success of these reactions. While many precursors of difluorocarbene have been described,<sup>12</sup> we have focused on silicon reagents with the general formula  $\text{Me}_3\text{SiCF}_2\text{X}$  (**4**)<sup>14</sup> (Scheme 4). The Ruppert–Prakash reagent is the most readily

## Scheme 4. Silicon Reagents

| $\text{Me}_3\text{SiCF}_2\text{X}$ | X  | reactivity | availability |
|------------------------------------|----|------------|--------------|
| <b>4a–d</b>                        |    |            |              |
| <b>a</b>                           | F  | *          | ****         |
| <b>b</b>                           | Cl | **         | **           |
| <b>c</b>                           | Br | ***        | ***          |
| <b>d</b>                           | I  | ****       | *            |



available, but it is the least reactive of these precursors. Though the generation of difluorocarbene from this silane may proceed upon treatment with sodium iodide at temperatures above 80 °C,<sup>15</sup> more forcing conditions may be required.<sup>16</sup> When considering both price and activity, (bromodifluoromethyl)-trimethylsilane (**4c**) is the reagent of choice. This silane can be prepared on a hundred-gram scale from the Ruppert–Prakash reagent by reduction<sup>17</sup> and bromination.<sup>18,19</sup> Currently, this reagent is commercially available from Aldrich. Chloro- and iodo-substituted silanes (**4b,d**) can be prepared from **4c** by reactions with chloride ion (for **4b**)<sup>20,21</sup> or by bromine/zinc exchange followed by iodination (for **4d**).<sup>22</sup>

Silane **4c** can generate difluorocarbene in the presence of weak, Lewis basic activators<sup>23</sup> (Scheme 5). For example,

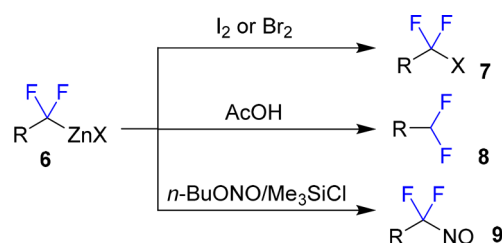
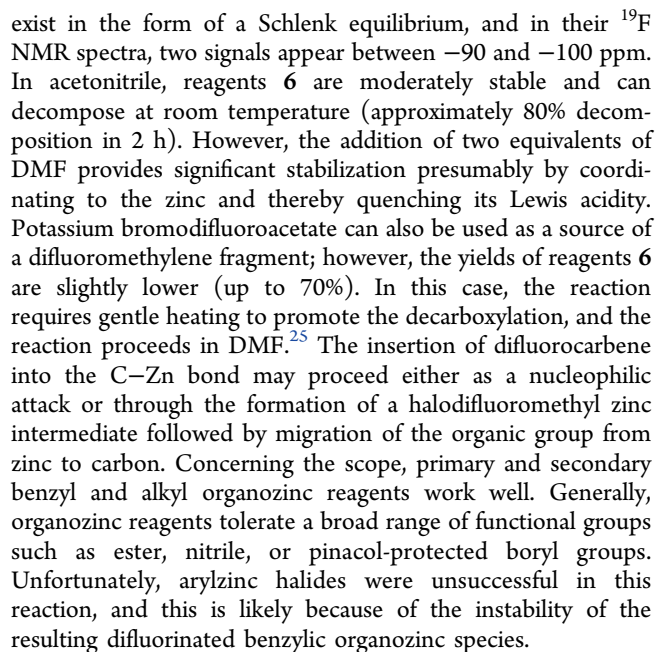
## Scheme 5. Generation of Difluorocarbene

| $\text{Me}_3\text{SiCF}_2\text{Br}$ ( <b>4c</b> ) | LB             | T, °C  |
|---------------------------------------------------|----------------|--------|
|                                                   | $\text{Br}^-$  | 80–110 |
|                                                   | HMPA, DMPU     | 0–rt   |
|                                                   | $\text{AcO}^-$ | –25 °C |

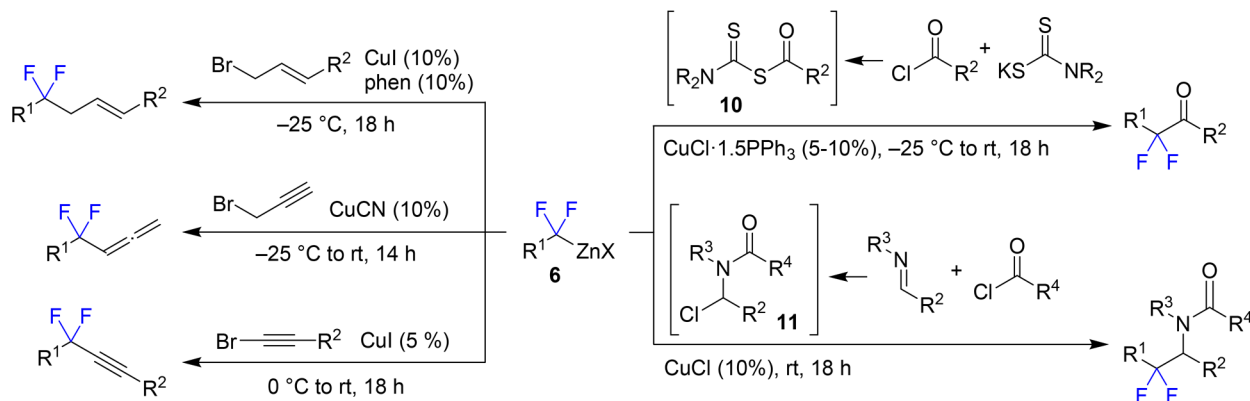
bromide ion can serve as a Lewis base from 80 to 110 °C.<sup>19</sup> Bromide can be used in catalytic amounts since it is formed upon fragmentation of the silane. The very weak bases hexamethylphosphoramide (HMPA) and *N,N'*-dimethylpropyleneurea (DMPU) can activate the silane at 0 °C, and these are probably the mildest conditions that can be used to generate difluorocarbene.<sup>20</sup> Our observations suggest that in the case of bromide ion and DMPU, the formation of the carbene is reversible, meaning that the combination of silane **4c** and a Lewis base can generate a small equilibrium concentration of difluorocarbene, which can react with a suitable reagent (nucleophile or alkene). Sodium acetate is an effective activator, and it can initiate the decomposition of **4c** at –25 °C in acetonitrile.<sup>11</sup>

## 3. ORGANOMETALLIC REAGENTS AS NUCLEOPHILES

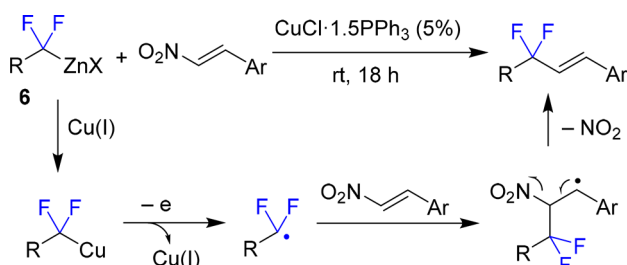
Organolithium and organomagnesium reagents are expected to react rapidly with difluorocarbene. However, due to the



Scheme 9. Copper-Catalyzed Cross-Couplings



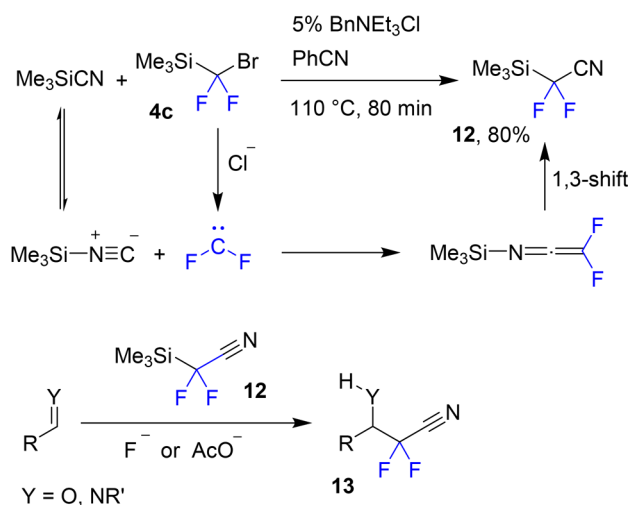
Scheme 10. Coupling of Reagents 6 with Nitrostyrenes



cross-coupling cycle but via a radical pathway. Thus, it is believed that, after the zinc/copper transmetalation, the organocopper intermediate is oxidized to generate a free radical, which adds to the double bond, and then nitrogen dioxide is eliminated.

In contrast to organozinc species, organoelement reagents with less polar carbon-element bonds are expected to be less reactive toward difluorocarbene. In particular, the reactivity of strong carbon–silicon bonds is intrinsically more moderate, and the direct insertion of difluorocarbene seems unlikely for kinetic reasons. It was rewarding to find that trimethylsilyl cyanide can react with silane **4c** in the presence of chloride ion for furnish silicon reagent **12**<sup>18</sup> (Scheme 11). We believe that the formal insertion of difluorocarbene into the Si–C bond proceeds via a stepwise mechanism. Silyl cyanide may exist in

Scheme 11. Synthesis of Reagent 12



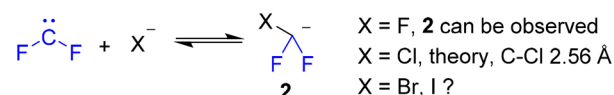
equilibrium with the isonitrile form, which can couple with difluorocarbene to generate a cumulene-type intermediate. Subsequent 1,3-shift of the silyl group from the nitrogen to silicon affords silane **12**. However, since no mechanistic evidence has been found, other pathways may be operative.

As a typical fluorinated silane, reagent **12** can be used for the fluoroalkylation of aldehydes and imines in the presence of fluoride or acetate anions.<sup>18</sup> Products **13** were employed as building blocks for the synthesis of various fluorine-substituted heterocyclic compounds.<sup>35</sup>

#### 4. HALIDE IONS AS NUCLEOPHILES

The interaction of difluorocarbene with a halide anion is a reversible process (Scheme 12). Moreover, either side of the

Scheme 12. Generation of Halodifluoromethyl Carbanion

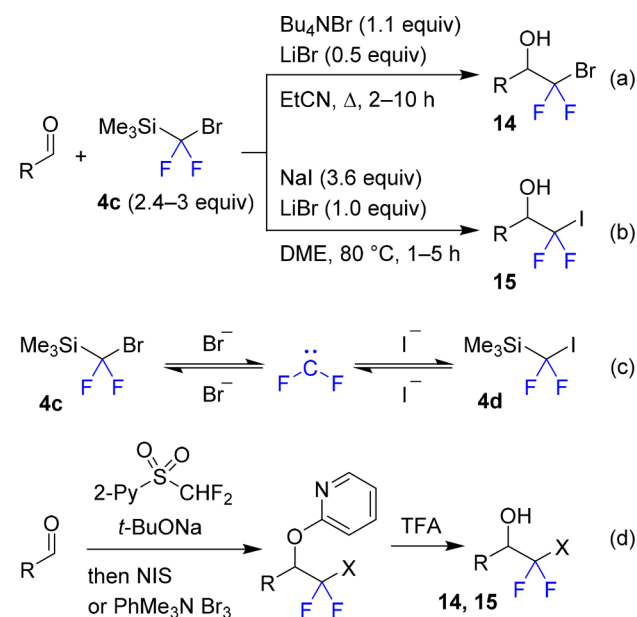


equilibrium can be accessed in the same system depending on the reaction conditions. For fluoride ( $X = F$ ), carbanion **2** (the  $CF_3$ -carbanion) can be directly observed in solution at low temperatures if a noncoordinating counterion is used.<sup>36</sup> In the presence of any Lewis acidic species (even potassium ion), the  $CF_3$ -carbanion decomposes with rupture of the C–F bond. In terms of synthesis, the use  $TMSCF_3$  activated by a Lewis base as a source of  $CF_3$ -carbanion for reactions with various electrophiles has become widely accepted.<sup>37</sup> Nevertheless, under certain conditions involving lithium or sodium cations, which have strong affinities for fluoride,  $TMSCF_3$  can also serve as a source of difluorocarbene.<sup>15</sup> For chloro-substituted silane,  $Me_3SiCF_2Cl$  (**4b**), both the carbene<sup>23</sup> and standard carbanionic<sup>38</sup> applications have been documented. The  $CF_2Cl$ -carbanion was evaluated by DFT, and a weakly bound structure with a long C–Cl bond distance of 2.56 Å was identified (for comparison, the C–Cl bond in **4b** was calculated to be 1.82 Å).<sup>23</sup> For bromide and iodide ions, the corresponding carbanions are expected to be highly unstable, and their trapping is challenging. Indeed, when a bromo-substituted silane ( $Me_3SiCF_2Br$ , **4c**) is combined with aldehydes in the presence of typical Lewis basic activators such as TBAF, no bromodifluoromethylation products are formed.

It was proposed that to shift the equilibrium shown in Scheme 12 toward carbanion **2** (or at least to generate a meaningful concentration of **2**), excess of halide ion should be

employed.<sup>22</sup> Indeed, the reaction of aldehydes with silane **4c** in the presence of tetrabutylammonium and lithium bromides at 100 °C afforded alcohols **14** after desilylative workup (Scheme 13, eq a). The method works only with aromatic and

Scheme 13. Halodifluoromethylation of Aldehydes



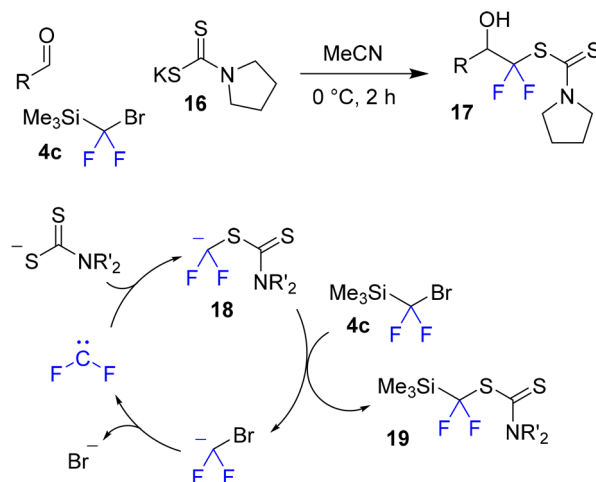
nonenolizable aliphatic aldehydes. The role of the lithium ion is merely to accelerate the reaction by Lewis acidic activation of the carbonyl group toward nucleophilic attack. The same protocol can be applied to the synthesis of iodo-substituted products employing iodinated silane **4d**.<sup>22</sup> To avoid the tedious preparation of silane **4d**, another method was developed based on the use of a combination of silane **4c** and sodium iodide<sup>39</sup> (eq b). In the presence of iodide, the halogen exchange equilibrium generates a mixture of silanes **4c** and **4d** (eq c). Sodium serves to scavenge bromide preferentially over iodide, thereby directing the reaction toward the formation of products **15**. Compounds **14** and **15** can also be obtained using a thiocarbonyl reagent under basic conditions followed by halogenation and removal of the pyridyl substituent<sup>40</sup> (eq d).

A combination of silane **4c** with aldehydes and potassium dithiocarbamate **16** leads to the formation of aldehyde addition products **17** bearing a dithiocarbamate fragment<sup>41</sup> (Scheme 14). In this reaction, carbanionic species **18** is generated by attack of the difluorocarbene by the dithiocarbamate anion. Anion **18** can add at the aldehyde or can undergo silylation to give silane **19**. In the absence of aldehyde, silane **19** can be isolated and characterized, and it can similarly be reacted with aldehydes leading to alcohols **17**.

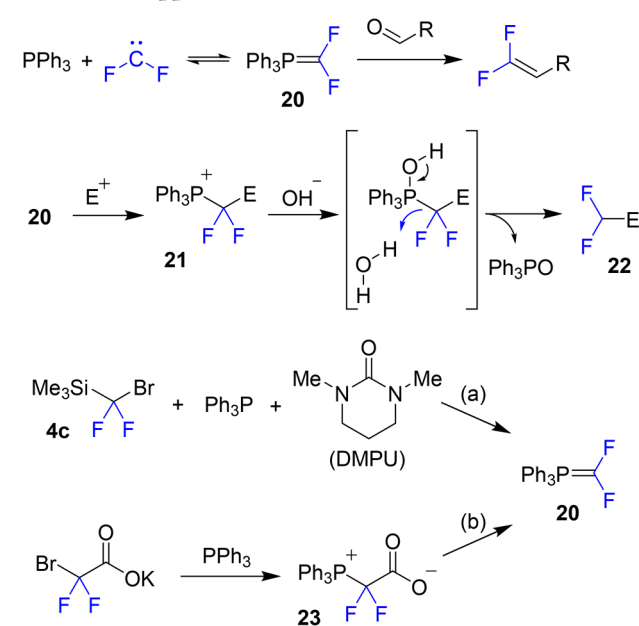
## 5. TRIPHENYLPHOSPHINE AS A NUCLEOPHILE

In 1964, it was reported that a combination of difluorocarbene generated from sodium chlorodifluoroacetate, triphenylphosphine and aldehyde affords *gem*-difluoroalkenes through an apparent Wittig reaction involving ylide **20**<sup>42</sup> (Scheme 15). In addition to chlorodifluoroacetate, other difluorocarbene precursors can be used in this process.<sup>43</sup> Ylide **20** has not been observed, but it exists in equilibrium with the phosphine and difluorocarbene.<sup>43</sup> We proposed to trap ylide **20** by an electrophile to generate phosphonium cation **21** with

Scheme 14. Synthesis of Dithiocarbamate-Substituted Difluorinated Alcohols



Scheme 15. Applications of Difluorinated Ylide **20**

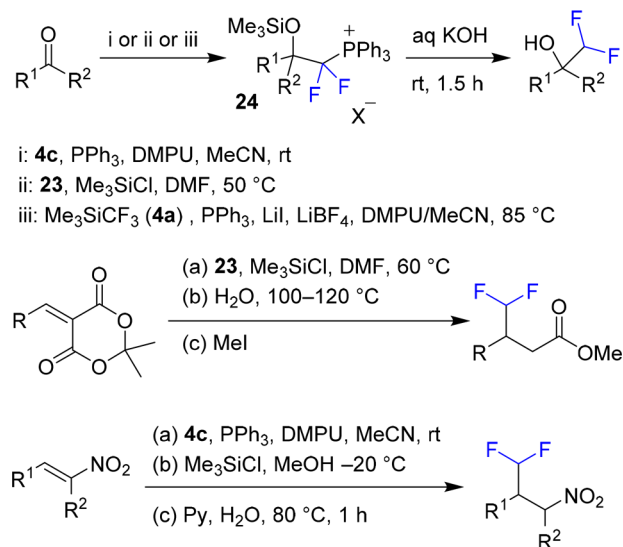


subsequent hydrolysis of the C–P bond to furnish product **22** with a difluoromethyl group. Though nonfluorinated phosphonium salts are not prone to facile cleavage of the C–P bond, salts **21** were expected to hydrolyze readily owing to the effect of the two fluorine atoms, which facilitate attack of the phosphorus by a hydroxide ion and favor heterolytic cleavage of the C–P bond.

Mild conditions to generate ylide **20** (0 °C to room temperature) involve the interaction of silane **4c** with the phosphine and DMPU (path a).<sup>44</sup> A cost-effective method for the preparation of ylide **20** based on betaine reagent **23** was developed.<sup>45</sup> This betaine can be obtained from potassium bromodifluoroacetate and triphenylphosphine and is an air-stable, crystalline compound, but upon mild heating (approximately 50 °C), it undergoes decarboxylation (path b).

Aldehydes and ketones can be difluoromethylated using silane **4c** to initially afford phosphonium salts **24**<sup>44</sup> (Scheme 16). Their protodephosphorylation can be achieved simply by addition of an aqueous base. For aldehydes, betaine reagent **23**

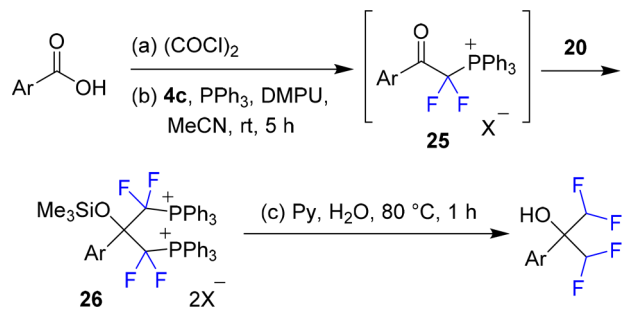
## Scheme 16. Nucleophilic Difluoromethylation Reactions



can be employed in combination with chlorotrimethylsilane.<sup>46</sup> It should be pointed out that the latter conditions are suitable not only for aromatic substrates but also for aliphatic enolizable aldehydes as well as some azomethines such as *N*-tosylimines, iminium ions, and boryl complexes derived from *N*-benzoylhydrazones. Michael acceptors derived from Meldrum's acid and nitro alkenes can also be successfully used as electrophiles to furnish nucleophilic difluoromethylation products.<sup>44,46</sup> Recently, it was demonstrated that even Ruppert-Prakash reagent (**4a**) in combination with triphenylphosphine and lithium salts can generate salts **24** from aldehydes<sup>47</sup> (Scheme 16, conditions iii). Though silane **4a** is readily available, the reaction can be accompanied by a nucleophilic trifluoromethylation of the carbonyl group, which is important for aldehydes containing an electron withdrawing group.

Acid chlorides derived from aromatic carboxylic acids combined with two equivalents of phosphobetaine **20** generated from the silane **4c**/DMPU system furnish tertiary alcohols bearing two difluoromethyl groups after dephosphorilation<sup>48</sup> (Scheme 17). It should be pointed out that the

## Scheme 17. Difluoromethylation of Carboxylic Acids

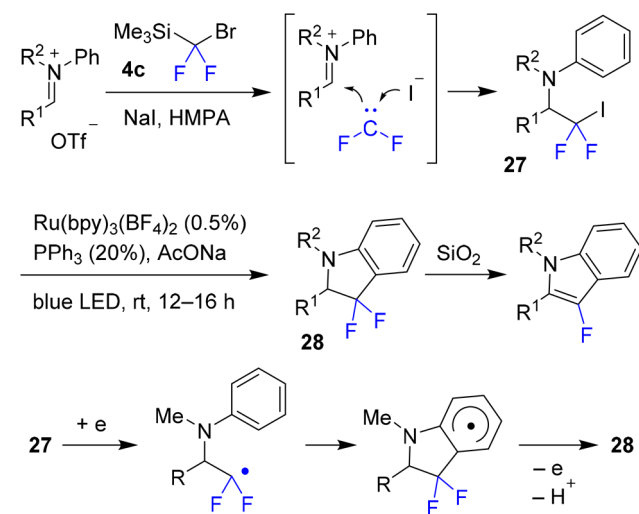


formation of double addition salts **26** is reversible, and heating these salts at 100 °C causes the elimination of 1 equiv of phosphobetaine **20** and provides salts **25**, which can be hydrolyzed to difluorinated ketones.

## 6. LIGHT-PROMOTED REACTIONS

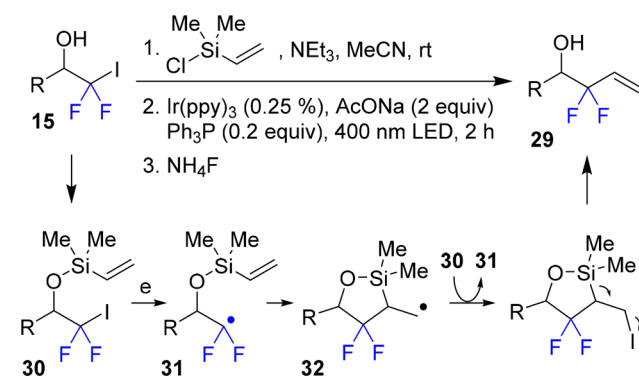
Perfluorinated organic iodides are known to serve as good substrates for photoredox reactions.<sup>49</sup> Electron withdrawing fluorine atoms favor single-electron reductions thereby providing convenient access to fluorinated radicals. In this regard, compounds with the CF<sub>2</sub>I-group, which can be accessed in a straightforward manner by the difluorocarbene methodology, can be subjected to further transformations by activation of the carbon–iodine bond. For example, nucleophilic iododifluoromethylation of *N*-aryl-substituted iminium ions using a combination of silane **4c** and sodium iodide leads to amines **27**<sup>20,50</sup> (Scheme 18). These amines undergo cyclization

## Scheme 18. Synthesis of 3-Fluoroindoles



in the presence of a ruthenium photocatalyst under irradiation with visible light to generate difluoroindolines **28**, and hydrogen fluoride is eliminated from these compounds on silica gel.<sup>50</sup> The reaction involves the single-electron reduction of amines **27** followed by intramolecular attack of the benzene ring by the fluorinated radical and subsequent oxidation and aromatization.

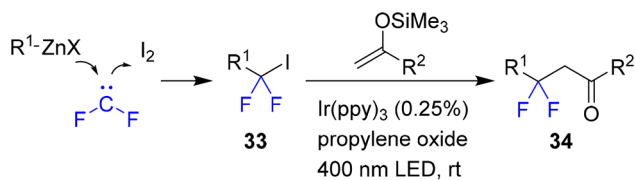
The iodine atom of alcohols **15** can be formally substituted by a vinyl group via a radical pathway (Scheme 19). Alcohols **15** are first silylated using chlorodimethyl(vinyl)silane leading to silyl ethers **30**. The presence of a photocatalyst and irradiation with 400 nm light-emitting diodes furnishes alcohols **29** after desilylation. It is likely that the reaction proceeds through radicals **31**, which undergo rapid 5-exo-dig cyclization.

Scheme 19. Vinylation of Alcohols **15**

In this case, a radical atom transfer chain mechanism was proposed.<sup>51</sup>

*gem*-Difluorinated iodides **33**, obtained from organozinc compounds and difluorocarbene, couple with silyl enol ethers under photoredox conditions<sup>52</sup> (Scheme 20). In this process,

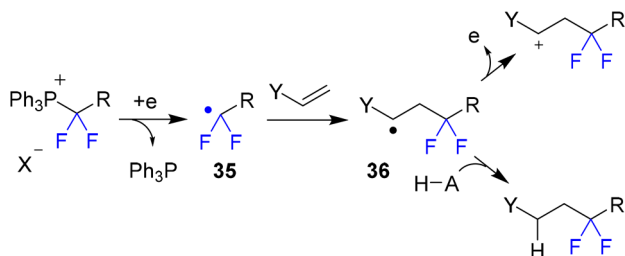
**Scheme 20. Fluoroalkylation of Silyl Enol Ethers**



propylene oxide is an important stoichiometric additive since it scavenges the iodotrimethylsilane formed as a byproduct. The reaction affords  $\beta,\beta$ -difluorinated ketones **34**, which would be difficult to obtain by other means owing to their propensity toward elimination of hydrogen fluoride.

Difluorinated phosphonium salts can also serve as sources of fluorinated radicals under photoredox conditions via the single-electron reduction of the carbon–phosphorus bond<sup>53,54</sup> (Scheme 21). Fluorinated radicals **35** can add to electron-rich

**Scheme 21. Activation of the C–P Bond**



and electron-poor alkenes, and the fate of the added radicals **36** depends on the nature of the alkene. When Y is an electron-donating substituent, radicals **36** can be oxidized to complete the photoredox cycle. If Y is electron-withdrawing, the oxidation of radicals **36** is not possible, and the radical can abstract a hydrogen atom from an external reagent.

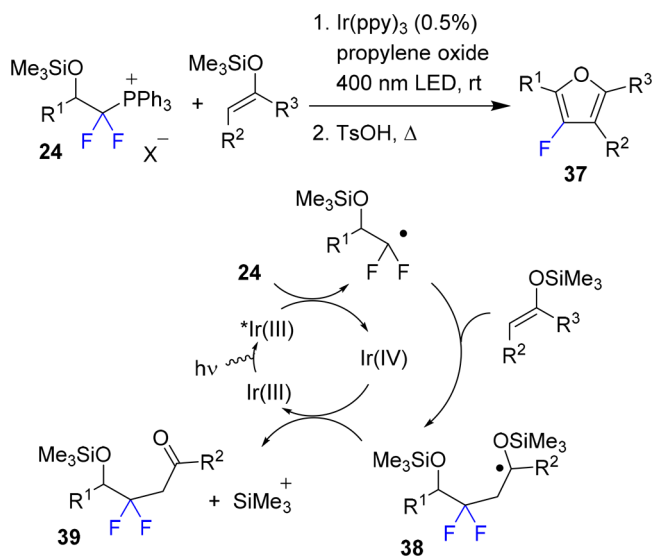
For example, phosphonium salts **24**, which are readily obtained in situ from carbonyl compounds, couple with silyl enol ethers in the presence of Ir(ppy)<sub>3</sub> (Scheme 22). The reaction likely proceeds via a classic photoredox cycle involving initial oxidative quenching of the excited Ir(III) catalyst followed by oxidation of radicals **38** by Ir(IV) with subsequent loss of the silyl group. Primary products **39** are cyclized into furans **37** by brief heating in the presence of *p*-toluenesulfonic acid.<sup>53</sup>

In the reactions of salts **24** with acrylonitrile, a Hantzsch ester can serve both as a source of electrons and as a source of hydrogen<sup>53</sup> (Scheme 23). Though the Hantzsch ester has an absorption maximum at approximately 360 nm, the shoulder of the absorption peak falls within the visible region, and therefore simple household fluorescent bulbs (CFLs) can be used to promote the reaction.

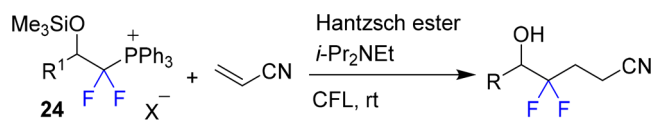
## 7. CONCLUSION

Despite being a short-lived intermediate, difluorocarbene is a useful building block for the synthesis of compounds bearing the difluoromethylene fragment. The electrophilic reactivity of

**Scheme 22. Synthesis of 3-Fluorofurans**



**Scheme 23. Hydrofluoroalkylation Reaction**



the carbene makes its reaction with nucleophiles straightforward. On the other hand, the poor stability of the resulting carbanionic species and facile reverse reaction back to the carbene complicate the selection of the appropriate reaction parameters.

The success of reactions based on difluorocarbene trapping is frequently determined by the source of difluorocarbene, and silicon reagents allow the mildest and most efficient systems. At the same time, phosphobetaine, originating from bromodifluoroacetate, can offer a cost-effective solution. Many challenges in the practical application of this difluorocarbene methodology still remain. For example, reactions of difluorinated phosphorus ylides with alkylating and arylating reagents remain to be developed. Applications of fluorinated phosphonium salts also require further investigation to broaden the scope of reagents capable of trapping fluorinated radicals via a photoredox cycle. For reactions involving phosphorus ylides, methods that are catalytic in phosphine are needed since this would increase the atom efficiency and would allow asymmetric versions of these fluoroalkylation processes.

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### Author Contributions

The manuscript was written through contributions of all authors.

### Notes

The authors declare no competing financial interest.

## Biographies

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